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Pipe coupling with pipe connection rings

Abstract

A pipe connection ring can include a plurality of segments and a plurality of deformable links. Each of the plurality of links can connect adjacent segments of the plurality of segments, the segments and links alternating about a circumference of the pipe connection ring, each pair of the adjacent segments and a corresponding link of the plurality of links connecting the adjacent segments and together with the adjacent segments defining a relief between the adjacent segments, each of the plurality of links configured to allow movement of the adjacent segments upon deformation of the link connecting the adjacent segments. The pipe connection ring can define a first axial end surface, a second axial end surface distal from the first axial end surface, an inner surface, and an outer surface distal from the inner surface.

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Background/Summary

REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 16/836,294, filed Mar. 31, 2020, which issued into U.S. Pat. No. 11,181,213 on Nov. 23, 2021, which is a continuation of U.S. application Ser. No. 15/378,138, filed Dec. 14, 2016, which issued into U.S. Pat. No. 10,641,419 on May 5, 2020, each of which is hereby specifically incorporated by reference herein in its entirety.

FIELD OF USE

(1) This disclosure relates to piping. More specifically, this disclosure relates to pipe couplings.

RELATED ART

(2) Pipe elements such as pipes, valves, and meters typically are not made of one piece. Rather, such pipe elements are formed in finite lengths and must be joined. One way of joining such pipe elements is with a pipe coupling. The pipe coupling typically includes a sealing member such as a gasket. Pipe couplings can be difficult to install, typically requiring the installation to both effectively seal a joint between the pipe elements and to prevent separation of the pipe elements against various internal or external forces.

SUMMARY

(3) It is to be understood that this summary is not an extensive overview of the disclosure. This summary is exemplary and not restrictive, and it is intended to neither identify key or critical elements of the disclosure nor delineate the scope thereof. The sole purpose of this summary is to explain and exemplify certain concepts of the disclosure as an introduction to the following complete and extensive detailed description.

(4) In one aspect, disclosed is a pipe system comprising: a first pipe element; a second pipe element; and a pipe coupling connecting an end of the second pipe element to an end of the first pipe element, the pipe coupling comprising: a first coupling segment defining an inner surface; a second coupling segment defining an inner surface, the inner surface of the first coupling segment and the inner surface of the second coupling segment defining: a first ring groove proximate to a first axial end of each of the first coupling segment and the second coupling segment, the inner surface of the first coupling segment and the inner surface of the second coupling segment comprising a transition portion on either axial side of the first ring groove, the first ring groove defining a groove bottom surface positioned radially outward from each of the transition portions; and a second ring groove proximate to a second axial end of each of the first coupling segment and the second coupling segment, the inner surface of the first coupling segment and the inner surface of the second coupling segment comprising a transition portion on either side of the second ring groove, the second ring groove defining a groove bottom surface positioned radially outward from each of the transition portions proximate to the second ring groove; a first pipe connection ring disposed within the first ring groove and engaged with the first pipe element, the first pipe connection ring comprising: an outer flange of the first pipe connection ring aligned with an axial direction of the first pipe connection ring and received within the first ring groove; and an inner flange of the first pipe connection ring connected to the outer flange and angled with respect to the outer flange; and a second pipe connection ring disposed within the second ring groove and engaged with the second pipe element, the second pipe connection ring comprising: an outer flange of the second pipe connection ring aligned with an axial direction of the second pipe connection ring and received within the second ring groove; and an inner flange of the second pipe connection ring connected to the outer flange of the second pipe connection ring and angled with respect to the outer flange of the second pipe connection ring.

(5) In a further aspect, also disclosed is a pipe coupling comprising: a first coupling segment defining an inner surface; a second coupling segment defining an inner surface, the inner surface of the first coupling segment and the inner surface of the second coupling segment defining: a first ring groove proximate to a first axial end of each of the first coupling segment and the second coupling segment, the inner surface of the first coupling segment and the inner surface of the second coupling segment comprising a transition portion on either side of the first ring groove, the first ring groove defining a groove bottom surface positioned radially outward from each of the transition portions; and a second ring groove proximate to a second axial end of each of the first coupling segment and the second coupling segment, the inner surface of the first coupling segment and the inner surface of the second coupling segment comprising a transition portion on either side of the second ring groove, the second ring groove defining a groove bottom surface positioned radially outward from each of the transition portions proximate to the second ring groove; a first pipe connection ring disposed within the first ring groove, the first pipe connection ring comprising: an outer flange of the first pipe connection ring aligned with an axial direction of the

first pipe connection ring and received within the first ring groove; and an inner flange of the first pipe connection ring connected to the outer flange and angled with respect to the outer flange; and a second pipe connection ring disposed within the second ring groove, the second pipe connection ring comprising: an outer flange of the second pipe connection ring aligned with an axial direction of the second pipe connection ring and received within the second ring groove; and an inner flange of the second pipe connection ring connected to the outer flange of the second pipe connection ring and angled with respect to the outer flange of the second pipe connection ring.

(6) In yet another aspect, also disclosed is a method of installing a pipe coupling to connect a first pipe element to a second pipe element, the method comprising: sliding a pipe coupling over an end of the first pipe element, the pipe coupling comprising: a first coupling segment defining an inner surface; a second coupling segment defining an inner surface, the inner surface of the first coupling segment and the inner surface of the second coupling segment defining: a first ring groove proximate to a first axial end of each of the first coupling segment and the second coupling segment, the inner surface of the first coupling segment and the inner surface of the second coupling segment comprising a transition portion on either side of the first ring groove, the first ring groove defining a groove bottom surface positioned radially outward relative to an axis of the pipe coupling from each of the transition portions; and a second ring groove proximate to a second axial end of each of the first coupling segment and the second coupling segment, the inner surface of the first coupling segment and the inner surface of the second coupling segment comprising a transition portion on either side of the second ring groove, the second ring groove defining a groove bottom surface positioned radially outward relative to an axis of the pipe coupling from each of the transition portions proximate to the second ring groove; a first pipe connection ring disposed within the first ring groove, the first pipe connection ring comprising: an outer flange of the first pipe connection ring aligned with an axial direction of the first pipe connection ring and received within the first ring groove; and an inner flange of the first pipe connection ring connected to the outer flange and angled with respect to the outer flange; and a second pipe connection ring disposed within the second ring groove, the second pipe connection ring comprising: an outer flange of the second pipe connection ring aligned with an axial direction of the second pipe connection ring and received within the second ring groove; and an inner flange of the second pipe connection ring connected to the outer flange of the second pipe connection ring and angled with respect to the outer flange of the second pipe connection ring; sliding the pipe coupling over an end of the second pipe element; and tightening the pipe coupling to engage the pipe connection ring with each of the first pipe element and the second pipe element.

(7) In yet another aspect, disclosed is a pipe connection ring comprising: a plurality of segments; and a plurality of deformable links, each of the plurality of links connecting adjacent segments of the plurality of segments, the segments and links alternating about a circumference of the pipe connection ring, each pair of the adjacent segments and a corresponding link of the plurality of links connecting the adjacent segments and together with the adjacent segments defining a relief between the adjacent segments, each of the plurality of links configured to allow movement of the adjacent segments upon deformation of the link connecting the adjacent segments; wherein the pipe connection ring defines a first axial end surface, a second axial end surface distal from the first axial end surface, an inner surface, and an outer surface distal from the inner surface.

(8) In yet another aspect, disclosed is a pipe connection ring comprising: a plurality of segments; and a plurality of deformable links, each of the plurality of links connecting adjacent segments of the plurality of segments, the segments and links alternating about a circumference of the pipe connection ring, the plurality of links and the plurality of segments defining an undulating or wave-like shape, each of the plurality of links configured to allow movement of the adjacent segments upon deformation of the link connecting the adjacent segments; wherein the pipe connection ring defines a first axial end surface, a second axial end surface distal from the first axial end surface, an inner surface, and an outer surface distal from the inner surface.

(9) In yet another aspect, disclosed is a pipe connection ring comprising: a plurality of segments, each of the plurality of segments defining in cross-section a beam comprising a plurality of legs; and a plurality of deformable links, each of the plurality of links connecting adjacent segments of the plurality of segments, the segments and links alternating about a circumference of the pipe connection ring, each of the plurality of links configured to allow movement of the adjacent segments upon deformation of the link connecting the adjacent segments; wherein the pipe connection ring defines a first axial end surface, a second axial end surface distal from the first axial end surface, an inner surface, and an outer surface distal from the inner surface.

(10) Various implementations described in the present disclosure may include additional systems, methods, features, and advantages, which may not necessarily be expressly disclosed herein but will be apparent to one of ordinary skill in the art upon examination of the following detailed description and accompanying drawings. It is intended that all such systems, methods, features, and advantages be included within the present disclosure and protected by the accompanying claims. The features and advantages of such implementations may be realized and obtained by means of the systems, methods, features particularly pointed out in the appended claims. These and other features will become more fully apparent from the following description and appended claims, or may be learned by the practice of such exemplary implementations as set forth hereinafter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate several aspects of the invention and together with the description, serve to explain various principles of the invention. The drawings are not necessarily drawn to scale. Corresponding features and components throughout the figures may be designated by matching reference characters for the sake of consistency and clarity.

(2) FIG. 1 is a perspective view of a pipe coupling in accordance with one aspect of the current disclosure.

(3) FIG. 2 is a sectional view of the pipe coupling of FIG. 1 taken along line 2-2 of FIG. 1.

(4) FIG. 3 is a perspective view of a pipe coupling in accordance with another aspect of the current disclosure.

(5) FIG. 4 is a perspective view of a coupling segment of the pipe coupling of FIG. 3 in accordance with one aspect of the current disclosure.

(6) FIG. 5 is a side view of the coupling segment of FIG. 4.

(7) FIG. 6 is a top view of the coupling segment of FIG. 4.

(8) FIG. 7 is a bottom view of the coupling segment of FIG. 4.

(9) FIG. 8 is a sectional view of the coupling segment of FIG. 4 taken along line 8-8 of FIG. 6.

(10) FIG. 9 is a sectional view of the coupling segment of FIG. 4 taken along line 9-9 of FIG. 5.

(11) FIG. 10A is a front view of a pipe connection ring in accordance with one aspect of the current disclosure.

(12) FIG. 10B is a detail view of the pipe connection ring of FIG. 10A taken from detail 10B of FIG. 10A.

(13) FIG. 11A is a front view of a pipe connection ring in accordance with another aspect of the current disclosure.

(14) FIG. 11B is a detail view of the pipe connection ring of FIG. 11A taken from detail 11B of FIG. 11A.

(15) FIG. 12A is a front view of a pipe connection ring in accordance with another aspect of the current disclosure.

(16) FIG. 12B is a detail view of the pipe connection ring of FIG. 12A taken from detail 12B of

FIG. 12A.

(17) FIG. 12C is a perspective view of the pipe connection ring of FIG. 12A.

(18) FIG. 13A is a front view of a pipe connection ring in accordance with another aspect of the current disclosure.

(19) FIG. 13B is a top view of the pipe connection ring of FIG. 13A.

(20) FIG. 13C is a detail view of the pipe connection ring of FIG. 13A taken from detail 13C of FIG. 13B.

(21) FIG. 13D is a perspective view of the pipe connection ring of FIG. 13A.

(22) FIG. 14A is a front view of a pipe connection ring in accordance with another aspect of the current disclosure.

(23) FIG. 14B is a detail view of the pipe connection ring of FIG. 14A taken from detail 14B of FIG. 14A.

(24) FIG. 14C is a sectional view of the pipe connection ring of FIG. 14A taken along line 14C-14C of FIG. 14B.

(25) FIG. 14D is a sectional view of the pipe connection ring of FIG. 14A taken along line 14C-14C of FIG. 14B in accordance with another aspect of the current disclosure.

(26) FIG. 15 is a perspective view of a pipe connection ring in accordance with another aspect of the current disclosure.

(27) FIG. 16 is a plan view of the pipe connection ring of FIG. 15 in a flattened and unformed condition.

(28) FIG. 17 is a perspective exploded view of a pipe system using the pipe connection ring of FIG. 15 in accordance with one aspect of the current disclosure.

(29) FIG. 18 is a sectional view of the pipe system of FIG. 17 in an assembled condition taken along line 18-18 in FIG. 17.

(30) FIG. 19A is a sectional view of a pipe system with the pipe coupling of FIG. 2 shown aligned with a first pipe element in accordance with another aspect of the current disclosure.

(31) FIG. 19B is a sectional view of the pipe system of FIG. 19A with the pipe coupling slid over an end of the first pipe element.

(32) FIG. 19C is a sectional view of the pipe system of FIG. 19A with a second pipe element aligned with the first pipe element.

(33) FIG. 19D is a sectional view of the pipe system of FIG. 19A with the pipe coupling slid over and aligned with both of the first pipe element and the second pipe element.

(34) FIG. 19E is a sectional view of the pipe system of FIG. 19A with the pipe coupling tightened around and connecting the first pipe element and the second pipe element.

(35) FIG. 20 is a sectional view of a pipe system with the pipe coupling tightened around and connecting the first pipe element and the second pipe element in accordance with another aspect of the current disclosure.

(36) FIG. 21 is a perspective view of a gasket for use in a pipe coupling such as the pipe coupling of FIG. 1 in accordance with another aspect of the current disclosure.

(37) FIG. 22 is a sectional view of the gasket of FIG. 21 taken along line 22-22 in FIG. 21.

(38) FIG. 23 is a sectional view of the gasket of FIG. 21 taken along line 23-23 in FIG. 21.

(39) FIG. 24 is a perspective view of a gasket for use in a pipe coupling such as the pipe coupling of FIG. 1 in accordance with yet another aspect of the current disclosure.

(40) FIG. 25 is a sectional view of the gasket of FIG. 24 taken along line 25-25 in FIG. 24.

(41) FIG. 26 is a sectional view of the gasket of FIG. 24 taken along line 26-26 in FIG. 24.

DETAILED DESCRIPTION

(42) The present invention can be understood more readily by reference to the following detailed description, examples, drawings, and claims, and their previous and following description.

However, before the present devices, systems, and/or methods are disclosed and described, it is to be understood that this invention is not limited to the specific devices, systems, and/or methods

disclosed unless otherwise specified, as such can, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting.

(43) The following description of the invention is provided as an enabling teaching of the invention in its best, currently known embodiment. To this end, those skilled in the relevant art will recognize and appreciate that many changes can be made to the one aspect of the invention described herein, while still obtaining the beneficial results of the present invention. It will also be apparent that some of the desired benefits of the present invention can be obtained by selecting some of the features of the present invention without utilizing other features. Accordingly, those who work in the art will recognize that many modifications and adaptations to the present invention are possible and can even be desirable in certain circumstances and are a part of the present invention. Thus, the following description is provided as illustrative of the principles of the present invention and not in limitation thereof.

(44) As used throughout, the singular forms “a,” “an” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a coupling segment” can include two or more such coupling segments unless the context indicates otherwise.

(45) Ranges can be expressed herein as from “about” one particular value, and/or to “about” another particular value. When such a range is expressed, another aspect comprises from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another aspect. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint.

(46) For purposes of the current disclosure, a material property or dimension measuring about X or substantially X on a particular measurement scale measures within a range between X plus an industry-standard upper tolerance for the specified measurement and X minus an industry-standard lower tolerance for the specified measurement. Because tolerances can vary between different materials, processes and between different models, the tolerance for a particular measurement of a particular component can fall within a range of tolerances.

(47) As used herein, the terms “optional” or “optionally” mean that the subsequently described event or circumstance may or may not occur, and that the description comprises instances where said event or circumstance occurs and instances where it does not.

(48) The word “or” as used herein means any one member of a particular list and also comprises any combination of members of that list.

(49) In one aspect, a pipe coupling used to join two pipe elements employs a housing comprising two approximately semicircular coupling segments joined together with nuts and bolts at each end. The housing is often used to secure two lengths of pipe together. In order to prevent leaking between the two pipes, a gasket is often secured around a joint created at the junction of the two pipes. The two coupling segments of the housing surround the gasket and press the gasket around the joint. It is contemplated that a pipe coupling such as that disclosed herein can have more than two coupling segments—for example and without limitation, three coupling segments, each occupying an arc length of approximately 120°; or four coupling segments, each occupying an arc length of approximately 90°. It is also contemplated that a pipe coupling such as that disclosed herein can have a single coupling segment that is able to slide over an end of a pipe element but which occupies an arc length of as much as 360° when tightened.

(50) Pipe couplings are often cast. Certain features of cast parts may vary in size from one part to another or have surfaces that are rougher or have material properties that are different than if the parts—or certain features of those parts—were processed using a different method such as forging, machining, or stamping. In one aspect, securing the coupling segments to end-facing pipe elements using specifically one or more separate pipe connection rings that are disclosed herein can yield benefits to those who install, service, and otherwise rely on the convenience and integrity of the

pipe connections.

(51) With previous pipe coupling designs, workers must sometimes follow a significant number of steps to install a pipe coupling. Workers must typically disassemble the entire apparatus and remove the gasket from the housing. Approximately half of the gasket is then forced over one end of one pipe. An end of the other pipe is forced into the remaining portion of the gasket, thereby joining the two pipes together. To secure the pipes, the two halves of the housing are placed around the gasket and then nuts and bolts or other fastening devices are used to complete the installation. Typically, the inner diameter of the gasket is slightly smaller than the outer diameter of the pipes. Therefore, the worker must stretch the gasket around the pipe ends.

(52) In one aspect, a pipe coupling and associated methods, systems, devices, and various apparatus are described herein that do not require the disassembly and assembly steps described immediately above. In one aspect, the pipe coupling comprises a pipe connection ring. In a further exemplary aspect, the pipe connection ring is compressible in a radial direction.

(53) In one aspect and as shown in FIGS. 1 and 2, the pipe coupling **100** includes new features not previously disclosed and/or eliminates other features. In one aspect, the pipe coupling **100** comprises a pair of coupling segments **200a,b**, a gasket **150**, a pair of pipe connection rings **300a,b**, and a pair of fastening devices **290a,b**. In one aspect, each of the coupling segments **200a,b** of the pipe coupling **100** comprises a first end **211a,b**, a second end **212a,b**, an outer surface **201a,b**, and an inner surface **202a,b** (shown in FIG. 2). In one aspect and as shown in FIG. 2, each of the outer surfaces **201a,b** is contoured to define a gently arched shape that is higher (or more radially outward) towards an axial midpoint of each coupling segment **200** and lower (or more radially inward) towards each of a pair of axial ends of each coupling segment **200**. Shown in an assembled but non-tightened state, the pipe coupling **100** of FIGS. 1 and 2 is shown with a noticeable gap **G** between the coupling segment **200a** and the coupling segment **200b**. In one aspect, the pipe connection rings **300a,b** are respectively positioned in or disposed within a first ring groove **250a** and a second ring groove **250b** (shown in FIG. 2) defined in the inner surfaces **202a,b** of the coupling segments **200a,b**. In one aspect, the first ring groove **250a** is proximate to a first axial end **205a,b** of the coupling segments **200a,b** and the second ring groove **250b** is proximate to a second axial end **206a,b** (shown in FIG. 2) of the coupling segments **200a,b**. In one aspect, the gasket is positioned in or disposed within a gasket groove **240** defined in the inner surfaces **202a,b** of the coupling segments **200a,b**. Each of the inner surfaces **202a,b** also defines a transition portion **272**, a transition portion **274**, a transition portion **276**, and a transition portion **278**. The transition portion **272** provides a transition between the first axial end **205a,b** and the first ring groove **250a**; the transition portion **274** provides a transition between the first ring groove **250a** and the gasket groove **240**; the transition portion **276** provides a transition between the gasket groove **240** and the second ring groove **250b**; and the transition portion **278** provides a transition between the second ring groove **250b** and the second axial end **206a,b**. In one aspect, each of the coupling segments **200a,b** comprises a first fastener pad **221a,b** and a second fastener pad **222a,b** proximate to the first end **211a,b** and the second end **212a,b**, respectively. Various detailed characteristics and functions of the pipe connection rings **300a,b** will be described in connection with the description of FIGS. **10A** through **14D**.

(54) In one aspect and as shown in FIG. 2, the gasket **150** comprises, for example and without limitation, a pair of sealing ridges **525a,b**, a central rib **530** defining sealing members **557a,b**, a deformation groove **517**, a pair of gasket channels **540a,b**—alternately described as pockets, the gasket channel **540a** proximate to the sealing ridge **525a** and the gasket channel **540b** proximate to the sealing ridge **525b**. In one aspect, the gasket **150** is received within the gasket groove **240** and each of the pipe connection rings **300a,b**, shown in simplified form for the sake of clarity, is respectively received within the ring grooves **250a,b**. In one aspect, in cross-section, the gasket **150** defines an overall height **570** measured in a radial direction from a radially outer surface **515** to a pair of sealing surfaces **526a,b** of the sealing ridges **525a,b**.

(55) In one aspect, the gasket **150** is ring-shaped (as shown in FIG. 17) with various axially outer drafted edges and axially inward drafted edges defined on the sealing ridges **525a,b**. The pair of sealing surfaces **526a,b** of the sealing ridges **525a,b**, respectively, are intended to contact pipe elements placed inside of the gasket **150** to provide a fluid seal for the pipe elements. The sealing surfaces **526a,b** face radially inwardly and extend substantially axially at rest, thereby being coannular with each other. Accordingly, in one aspect, the shape of each of the sealing surfaces **526a,b** approximates a cylinder that is about coaxial with the pipe elements intended to be used with the pipe coupling **100**. Thus, the sealing surfaces **526a,b** appear as lines that are parallel with an axial direction of the gasket **150** and an axial direction of the pipe elements **110,120** in cross-sectional view, as seen in FIGS. 19A-19E. Any angle with respect to the pipe elements **110,120** is minimal. The axial ends of each sealing surface **526a,b** may be rounded or chamfered or otherwise shaped to ease the insertion of pipe elements **110,120**.

(56) The gasket **150** may be made of rubber, plastic, cork, wood, metal, ceramic, polymer, elastomer, rosin, foam, any combination of the foregoing materials, or any material suitable for sealing two pipe elements joined in end-to-end relationship. A “pipe element” can include, for example and without limitation, pipes, valves, meters, fittings, or any other piping structure suitable to be sealed.

(57) The gasket **150** defines the gasket channels **540a,b**. The gasket channels **540a,b** are pockets into which fluid media may flow when the gasket **150** is in use. The gasket channels **540a,b** are tubular channels in the current embodiment but may be various shapes in various embodiments. When placed in sealing contact with an exterior surface of a pipe element, the gasket channels **540a,b** allow some fluid pressure to aid in sealing the sealing ridges **525a,b** against the pipe elements **110,120** by exerting pressure against an axially outer surface of the sealing ridges **525a,b**, although such use is not necessary for successful sealing of the gasket **150**. The central rib **530** decreases in thickness from a radially outermost portion to its termination radially inward.

(58) In addition, when the gasket **150** is in use, the sealing members **557a,b** act to prevent substantial fluid media flow into the gasket channels **540a,b**. When placed in sealing contact with exterior surfaces of pipe elements, a pair of sealing surfaces **532a,b** of the sealing members **557a,b** prevent substantial fluid media flow into gasket channels **540a,b**, retaining fluid media flow in a groove between the sealing members **557a,b**. The gasket, in alternative embodiments, may comprise a plurality of central ribs, each with at least one sealing member and at least one sealing surface, which perform the same function as described above to prevent substantial media flow into the gasket channels. One problem that the central rib **530** can alleviate is the buildup of fluids in the gasket channels **540a,b**. For example, in applications where fluid media is water in cold temperature environments, preventing water buildup in the gasket channels **540a,b** can lead to damage to the gasket **150** if the water freezes and, thereby, expands.

(59) In one aspect and as shown in FIG. 3, the pipe coupling **100**—including the coupling segments **200a,b**—comprises new features not previously disclosed and/or eliminates other features. In the current aspect, the coupling segments **200a,b** respectively define outer surfaces **201a,b** and inner surfaces **202a,b** that differ in various aspects to the outer surfaces **201a,b** and inner surfaces **202a,b** of the coupling segments **200a,b** of FIGS. 1 and 2. In particular, the gently arched shape of the outer surfaces **201a,b** is broken towards each of the axial ends **205,206** of each coupling segment **200** by a plurality of edge treatments **260a,b** corresponding to a plurality of groove bottom segments **270** shown in FIGS. 7-8. In one aspect, the edge treatments **260a,b** are planar surfaces that help maintain a more constant thickness of the coupling segment in portions of the coupling segment defining the ring grooves **250a,b**. In one aspect, this ensures that the pipe coupling does not become too thin in any one area. In another aspect, this reduces the amount of material used to form each of the coupling segments **200a,b** to only the amount required the minimum thickness. In another aspect, a more constant thickness improves the manufacturability of the coupling segments **200a,b** by facilitating more even cooling of a coupling segment after it has been cast or molded. As

will be described in connection with the description of FIGS. 12A-12C, each of the pair of pipe connection rings **300a,b** of FIG. 3 exhibits characteristics that differ from the characteristics of the pipe connection rings **300a,b** of FIGS. 1 and 2.

(60) In various aspects and as shown in FIGS. 4-9, the coupling segment **200** respectively defines fastener openings **224,225** in fastener pads **221,222** proximate to the first end **211** and the second end **212**. In one aspect, the fasteners opening **224,225** may be capsule-shaped and extend in an axial direction defined by the pipe coupling **100**, though the fastener openings may have various other shapes in other aspects, including a circular or rectangular shape, and can extend in a direction orthogonal to the axial direction or other directions as required for particular applications. In one aspect, to facilitate the manufacture of the coupling segment **200**, an inner surface of the fastener openings **224,225** are drafted as shown in FIG. 5 and other surfaces shown in FIGS. 4-9 of the coupling segment **200** are drafted as shown. In one aspect, each of the fastener pads **221,222** tapers toward the first end **211** and the second end **212** as shown in FIGS. 6 and 7. In one aspect, the gasket groove **240** is centered axially in the coupling segment **200**. In one aspect, each of the ring grooves **250a,b** is positioned axially outward from the gasket groove **240** as shown in FIGS. 7 and 9. In one aspect, the groove bottom segments **270**—which can also be described as flat portions of a groove bottom surface defined in the inner surface **202** of the coupling segment **200**—of the ring grooves **250a,b** are each flat in cross-section as shown in FIG. 8. In one aspect, the groove bottom segments **270** of the ring grooves **250a,b** are flat in cross-section and parallel to an axial direction of the coupling segment in cross-section as shown in FIG. 9. In one aspect, the groove bottom segments **270** form a substantially half-polygonal shape in a radially oriented plane. In one aspect where a pair of coupling segments **200** is assembled in a pipe coupling **100**, the groove bottom segments **270** of both coupling segments **200** form a ten-sided polygonal profile. In another aspect, the total number of groove bottom segments **270** can be greater than or less than ten.

(61) In one aspect, when the coupling segments **200a,b** are formed with the groove bottom segments **270**, an outer surface **304** (shown in FIG. 10A) of the substantially circular pipe connection rings **300a,b** can be configured to contact only a portion of the ring grooves **250a,b**. For example and without limitation, the outer surface **304** of each of the pipe connection rings **300a,b** can be configured to contact a portion of the ring grooves **250a,b** of each of the coupling segments **200a,b** only at respective midpoints of the coupling segments **200a,b** between the respective first end **211a,b** and the second end **212a,b**. In one aspect, the groove bottom segments **270** enable portions of an outer surface **304** of the pipe connection ring **300** to slide more easily during tightening of the pipe coupling **100** while the pipe connection ring **300** compresses due to the shape and orientation of the groove bottom segments **270** and the reduced contact surface area between the outer surface **304** of the pipe connection ring **300** and the groove bottom segments **270**. The ring grooves **250a,b** may also have a radiused profile instead of the polygonal profile shown.

(62) In various aspects and as shown in FIGS. 10A-14D, the pipe connection ring **300** comprises a plurality of segments and a plurality of links that are deformable with respect to the segments. In various aspects, each of the plurality of links connects a one of the plurality of segments to another of the plurality of segments. In various aspects, the segments and the links alternate about the circumference of the pipe connection ring **300**. In one aspect, the pipe connection ring **300** is a closed ring, wherein to be a “closed ring” means to have a ring without ends or to have a ring where the two ends are joined or connected to one another. If a ring has two ends that are “joined” or “connected,” however, the two ends cannot be simply touching each other such that the ring could be made to open up a gap such as in the case of a split ring.

(63) In one aspect and as shown in FIGS. 10A and 10B, the pipe connection ring **300** comprises a plurality of alternating segments **310** and segments **320**, connected to each other by a plurality of links **330** positioned therebetween. In one aspect, each segment **310** comprises an inner side **311**, an outer side **312**, a first end **313**, and a second end **314** and defines a relief **315** in the inner side

311 proximate to a first linking end **316** of the segment **310** and a relief **317** in the outer side **312** proximate to a second linking end **318** of the segment **310**. In one aspect, the segment **320** comprises an inner side **321**, an outer side **322**, a first end **323**, and a second end **324**. In one aspect, the segment **310** and the segment **320** together define a relief **325** proximate to a first linking end **326** of the segment **320** and a relief **327** proximate to a second linking end **328** of the segment **320**. In one aspect, each of the reliefs **315,317** comprises, for example and without limitation, a semicircular shape that helps define the shape of the link **330** such that the link **330** can have a cross-sectional area that is smaller than a cross-sectional area of the segment **310** or the segments **320** and will therefore deform first and also deform predictably—specifically, at the point where the cross-sectional area of the link **330** is at a minimum. The reliefs **315,317** may include a V-shaped portion. In one aspect, each of the reliefs **325,327** comprises opposing sides that are angled with respect to one another by a relief angle **A1**. In various aspects, deformation of the pipe connection ring **300** can cause the relief angle **A1** of the reliefs **325,327** to decrease from its original value or can otherwise cause the shape of the reliefs **315,317,325,327** to be altered. In various aspects, each of the first ends **313,323** and the second ends **314,324** are angled by a slant angle **A4** with respect to a tangent line defined by a circumferential axis **307** of the pipe connection ring **300**. In one aspect, the pipe connection ring **300** defines a first axial end surface **301**, a second axial end surface distal from the first axial end surface **301**, an inner surface **303**, and the outer surface **304**.

(64) In one aspect, the link **330** between each pair of adjacent segments such as between any two adjacent segments **310,320** is a small portion of material connecting the first segment **310** to the second segment **320** and is deformable with respect to the segments **310,320** to allow the pipe connection ring **300** to be compressed in a radial direction. In one aspect, by being weaker than the segments **310,320**, the link **330** results in predictable and constant compression of the pipe connection ring **300** in a radial direction at each point around the pipe connection ring **300**. In one aspect, the plurality of segments defines a local inner radius **1081** and a local outer radius **1082** centered about a point that is offset by an offset distance **1060** from a central axis **305** of the pipe connection ring **300**. In one aspect, adjacent links **330** are positioned at regular intervals around the circumference of the pipe connection ring **300**. In one aspect, the adjacent links **330** connected to the linking ends **326,328** of the segment **320** are spaced apart as shown by a spacing angle **A2** and the adjacent links **330** connected to the linking ends **316,318** of the segment **310** are spaced apart as shown by a spacing angle **A3**. In one aspect, the spacing angle **A5** represented by the sum of the spacing angle **A2** and the spacing angle **A3** is equal to, for example and without limitation, approximately 22.5 degrees due to the existence of sixteen segment pairs, each segment pair containing a segment **310** and a segment **320**.

(65) The pipe connection ring **300** shown in FIGS. **10A** and **10B** and other exemplary pipe connection rings **300** such as those shown in FIGS. **11A-14D** can be formed from a flat blank of raw material such as, for example and without limitation, carbon steel and stainless steel including spring steel grades of each. The pipe connection ring **300** shown can be formed using one or more material removal processes such as, for example and without limitation, machining, stamping, punching, laser-cutting, abrasive-water-jet-cutting, and chemical milling or etching, optionally in combination with forming processes such as, for example and without limitation, casting, forging, stamping, bending, and three-dimensional printing. Likewise, in one aspect, the plurality of segments **310,320** and the plurality of links **330** define an axial ring thickness **308** (i.e., a thickness of the pipe connection ring **300** in an axial direction) shown in FIG. **13B** that is constant between the pipe connection ring **300** as shown and the pipe connection ring **300** as compressed in a radial direction (i.e., when the pipe coupling **100** is in an assembled and tightened state). The axial ring thickness **308**, however, need not be constant in every embodiment of the pipe connection ring **300**.

(66) During installation of the pipe coupling **100**, the pipe connection ring **300** can be compressed in a radial direction towards the central axis **305** by gradual tightening of the pipe coupling **100**

with the pipe connection ring **300** inside a ring groove **250** of the pipe coupling **100**. During radial compression of the pipe connection ring **300**, the center of the local inner radius **1081** and the local outer radius **1082** of the respective datum surfaces of each segment **310,320** of the pipe connection ring **300** approaches the central axis **305** until the center of each of the radii **1081,1082** is coaxial with the central axis **305**. In one aspect, during radial compression of the pipe connection ring **300** or between a non-assembled state and an assembled and tightened state of the pipe coupling **100**, each of the plurality of segments **310,320** rotates in one of a direction **R1** and a direction **R2** with respect to a circumferential axis **307** of the pipe connection ring **300**. In one aspect, it is the alternating placement of the links **330** on the inside and outside of the pipe connection ring **300** that causes this rotation. This rotation may be difficult to measure absent fine measuring techniques because of the relatively small size of the offset distance **1060** relative to the radii **1081,1082** and an overall inner radius **1091** and an overall outer radius **1092** as measured from the central axis **305**. In one aspect, each of the plurality of segments **310** rotates with respect to a circumferential axis **307** of the pipe connection ring **300** between a non-assembled state and an assembled and tightened state. The local inner radius **1081** may be made to match the overall inner radius **1091**, however, and the local outer radius **1082** may be made to match the overall outer radius **1092**.

(67) In another aspect and as shown in FIGS. **11A** and **11B**, the pipe connection ring **300** comprises the plurality of segments **310** and the plurality of links **330** positioned therebetween; however, each of the plurality of segments **310** and the plurality of links **330** are of a different shape than previously described. In one aspect, each segment **310** comprises the inner side **311**, the outer side **312**, the first end **313**, and the second end **314** and defines a relief **315** in the first end **313** proximate to the first linking end **316** and an relief **317** in the second end **314** proximate to the second linking end **318**.

(68) In one aspect, the link **330** between each pair of adjacent segments **310** is a small portion of material connecting one segment **310** to an adjacent segment **320** and is deformable with respect to the adjacent segments **310** to allow the pipe connection ring **300** to be compressed in a radial direction. The link **330** comprises legs **1110a,b** and a hairpin bend **1120**. In one aspect, the leg **1110a** is connected to the second linking end **318** of one segment **310** and the leg **1110b** is connected to the first linking end **316** of an adjacent segment **310**. In one aspect, the legs **1110a,b** comprise opposing sides that are angled with respect to one another by the relief angle **A1**. In various aspects, deformation of the pipe connection ring **300** can cause the relief angle **A1** of the reliefs **325,327** to decrease from its original value or can otherwise cause the shape of the reliefs **315,317,325,327** to be altered. In one aspect, by being weaker than the segments **310**, the link **330** results in predictable and constant compression of the pipe connection ring **300** in a radial direction at each point around the pipe connection ring **300**. In one aspect, the plurality of segments defines the local inner radius **1081** and the local outer radius **1082** centered about a point that is offset by the offset distance **1060** from the central axis **305** of the pipe connection ring **300**. In one aspect, the plurality of segments **310** and the plurality of links **330** define a radial ring thickness **306** that is constant between a non-assembled state and an assembled and tightened state.

(69) In one aspect, adjacent links **330** are positioned at regular intervals and spaced, for example and without limitation, approximately twenty degrees apart as shown by the spacing angle **A2**. In one aspect, the pipe connection ring **300** thus comprises eighteen different segments **310**. In one aspect, each of the plurality of segments **310** remains aligned with (i.e., does not rotate with respect to) a circumferential axis **307** of the pipe connection ring **300** between a non-assembled state and an assembled and tightened state. In one aspect, the pipe connection ring **300** is in an uncompressed condition when the pipe coupling **100** is in an untightened position, and the pipe connection ring **300** is in a compressed condition when the pipe coupling **100** is in a tightened position.

(70) In another aspect and as shown in FIGS. **12A** through **12C**, the pipe connection ring **300** comprises the plurality of segments **310** and the plurality of links **330** positioned therebetween; however, each of the plurality of segments **310** and the plurality of links **330** are of a different

shape than previously described. In one aspect, the segment **310** has the shape of a bar of constant thickness aligned along an axial direction, the segment **310** including the inner side **311**, the outer side **312**, the first linking end **316**, and the second linking end **318**. As shown in FIGS. **12A** through **12C**, the plurality of segments **310** and the plurality of links **330** defines an undulating or wave-like shape that undulates between the overall inner radius **1091** and the overall outer radius **1092**. As shown, the links **330** are bends in the material. In one aspect, an internal radius **1210** of an axially aligned edge of the pipe connection ring **300** is sized such that compression in a radial direction of the pipe connection ring **300** can cause deformation primarily at the link **330** due to the increased stress concentration in the pipe connection ring **300** at each of the links **330** due to the small size of the internal radius **1210**. In one aspect, an external radius **1220** of an axially aligned edge of the pipe connection ring **300** is, for example and without limitation, greater than the internal radius **1210** by an amount equal to a thickness of the material forming the pipe connection ring **300**, which in one aspect is equal to a distance from the inner side **311** to the outer side **312** of the segment **310**. The external radius **1220** could also be greater than the internal radius **1210** by an amount greater than a nominal thickness of the material forming the pipe connection ring **300**, causing an effective thickness of the link **330** to decrease. Reducing the effective thickness of the link **330** relative to the thickness of the segment **310** causes the link **330** to deform more easily and still have the rigidity provided by the segments **310**. In one aspect, the plurality of segments **310** and the plurality of links **330** define a radial ring thickness **306** that increases between a non-assembled state and an assembled and tightened state.

(71) In one aspect, adjacent links **330** are spaced, for example and without limitation, approximately five degrees apart as shown by each of the spacing angles **A2**, **A3**. In one aspect, the pipe connection ring **300** thus comprises thirty-six different segments **310**. In one aspect, each of the plurality of segments **310** is inclined with respect to and continues to rotate in one of the direction **R1** and the direction **R2** with respect to a circumferential axis **307** of the pipe connection ring **300** between a non-assembled state and an assembled and tightened state. In one aspect, the plurality of segments **310** and the plurality of links **330** define undulations in the radially outward facing outer surface **304** of the pipe connection ring **300**.

(72) In another aspect and as shown in FIGS. **13A** through **13D**, the pipe connection ring **300** comprises the plurality of segments **310** (shown in FIG. **13B**) and the plurality of links **330** (also shown in FIG. **13B**) positioned therebetween; however, each of the plurality of segments **310** and the plurality of links **330** are of a different shape than previously described. In one aspect, the segment **310** has the shape of a bar of constant thickness aligned in a radial direction, the segment **310** including the inner side **311** (shown in FIG. **13D**), the outer side **312** (also shown in FIG. **13D**), the first linking end **316** (shown in FIG. **13C**), and the second linking end **318** (also shown in FIG. **13C**). As shown in FIGS. **13A** through **13D**, the plurality of segments **310** and the plurality of links **330** defines an undulating or wave-like shape that undulates between an axially outermost portion of the first axial end surface **301** and an axially outermost portion of the second axial end surface **302**. As shown, the links **330** are bends in the material. In one aspect, the internal radius **1210** (shown in FIG. **13C**) of a radially aligned edge of the pipe connection ring **300** is sized such that compression in a radial direction of the pipe connection ring **300** can cause deformation at the link **330** due to the increased stress concentration in the pipe connection ring **300** at each of the links **330** due to the small size of the internal radius **1210**. In another aspect, the segments **310** can also deform during radial compression of the pipe connection ring **300**. In one aspect, the external radius **1220** of a radially aligned edge of the pipe connection ring **300** is, for example and without limitation, greater than the internal radius **1210** by an amount equal to a thickness of the material forming the pipe connection ring **300**, which in one aspect is equal to a distance from the first axial end surface **301** to the second axial end surface **302** of the segment **310**.

(73) In one aspect, adjacent links **330** are spaced, for example and without limitation, approximately five degrees apart as shown by each of the spacing angles **A2**, **A3**. In one aspect, the

pipe connection ring **300** thus comprises thirty-six different segments **310**. In one aspect, each of the plurality of segments **310** is inclined with respect to and continues to rotate in one of the direction **R3** and the direction **R4** with respect to a diameter of the pipe connection ring **300** between a non-assembled state and an assembled and tightened state. In one aspect, the plurality of segments **310** and the plurality of links **330** define an axial ring thickness **308** that increases from a non-assembled state to an assembled and tightened state. In one aspect, an increase in the axial ring thickness **308** during radial compression of the pipe connection ring **300** can promote more constant deformation of the pipe connection ring **300** about its circumference. For example and without limitation, any portion of the pipe connection ring **300** which tends to deform more quickly during radial compression can approach or contact the sides of the ring groove **250**. If the pipe connection ring **300** contacts the ring groove **250**, deformation can slow or stop until other portions of the pipe connection ring **300** deform to an equal degree or until compression of the pipe connection ring **300** stops due to the pipe connection ring **300** engaging the pipe element **110,120**. In one aspect, the plurality of segments **310** and the plurality of links **330** define undulations in the axially outward facing axial end surfaces **301,302** of the pipe connection ring **300**.

(74) In another aspect and as shown in FIGS. **14A** through **14D**, the pipe connection ring **300** comprises the plurality of segments **310** and the plurality of links **330** positioned therebetween; however, each of the plurality of segments **310** and the plurality of links **330** are of a different shape than previously described.

(75) In one aspect as shown in FIG. **14C**, the segment **310** has the cross-sectional shape of a beam in the shape of a letter “S” of constant thickness aligned substantially along a circumferential axis **307**, the cross-sectional shape of the segment **310** including an inner leg **1411**, an outer leg **1412**, a first end leg **1413** extending from the inner leg **1411**, a second end leg **1414** extending from the outer leg **1412**, and a cross leg **1415** extending diagonally between the inner leg **1411** and the outer leg **1412**. In another aspect shown in FIG. **14D**, the first end leg **1413** comprises a portion that is angled from the first axial end surface **301** and the second end leg **1414** comprises a portion that is angled from the second axial end surface **302** by a bend angle **A7**.

(76) As shown in FIGS. **14A** through **14D**, the plurality of segments **310** and the plurality of links **330** defines an undulating or square wave-like shape that undulates between the overall inner radius **1091** and the overall outer radius **1092**. In one aspect, compression of the pipe connection ring **300** in a radial direction will cause deformation primarily at the link **330** because the cross-sectional area of the pipe connection ring **300** is at a minimum at each of the links **330**. Each of the plurality of links **330** may be positioned alternately proximate to an inner diameter and proximate to an outer diameter of the pipe connection ring **300**.

(77) In one aspect, adjacent links **330** are spaced, for example and without limitation, approximately 4.5 degrees apart as shown by each of the spacing angles **A2,A3**. In one aspect, the pipe connection ring **300** thus comprises forty different segments **310**. In one aspect, each of the plurality of segments **310** is inclined with respect to and continues to rotate in one of the direction **R1** and the direction **R2**—or in a direction opposite of the direction **R1** or the direction **R2**—with respect to a circumferential axis **307** of the pipe connection ring **300** between a non-assembled state and an assembled and tightened state. The alternating placement of the links **330** on the inside and outside of the pipe connection ring **300** causes this rotation. In one aspect, the pipe connection ring **300** is allowed to not only compress radially such that the overall inner radius **1091** and the overall outer radius **1092** are decreased, but the pipe connection ring **300** is also allowed to compress radially such that an overall outer radius **1092** will decrease more than an overall inner radius **1091**. In various aspects, because of the existence of clearance gaps **1420a,b**, a cross-section of the pipe connection ring such as shown in FIG. **14C** or **14D** can be “crushed” such that the radial ring thickness **306** of the pipe connection ring decreases. In various aspects, a cross-sectional geometry of the pipe connection ring **300** can be modified to increase or decrease the tendency of the pipe connection ring to “crush” by making modifications that increase or decrease the moment of inertia

of the cross-sectional geometry. In FIGS. **10A** through **14D**, each pipe connection ring **300** is shown in an uncompressed state. In a compressed state, each pipe connection ring **300** will have a similar shape but will have a smaller diameter due to the plurality of segments **310,320** coming closer to one another.

(78) In another aspect and as shown in FIG. **15**, the pipe connection ring **300** comprises new features not previously disclosed and/or eliminates other features. In one aspect shown, the pipe connection ring **300** comprises an outer flange **1510** aligned with an axial direction of the pipe connection ring **300**, a radial flange **1520** aligned with a radial direction of the pipe connection ring **300**, and an inner flange **1530**—which can be serrated as shown—angled with respect to an axial direction of the pipe connection ring **300** by an incline angle **A6**. In one aspect, each of the segments **310** has a cross-sectional shape that is aligned substantially along a circumferential axis. In one aspect, the inner flange **1530** of the pipe connection ring **300** defines a plurality of relief cuts **1535** and ends **1537**.

(79) As shown in FIG. **16**, a flattened version of the pipe connection ring **300** before it is formed defines a pair of bend lines **1610a,b**, where the bend line **1610a** is positioned between the portion of the pipe connection ring **300** forming the outer flange **1510** and the bend line **1610b** is positioned between the portion of the pipe connection ring **300** forming the radial flange **1520** and the portion of the pipe connection ring **300** forming the inner flange **1530**. In one aspect, radial expansion or bending of an inner flange **1530** in a radially outward direction and the accompanying increase in an overall inner diameter equaling twice the overall inner radius **1091** of the pipe connection ring **300** in a radial direction during the installation of the pipe coupling **100** will allow the pipe connection ring **300** to slide over the pipe elements **110,120** and then return to an unexpanded condition as shown in FIG. **18**. Such a pipe connection ring **300** need not be compressed radially to function as intended in the pipe coupling **100** shown.

(80) In one aspect, relief cuts **1535** are spaced, for example and without limitation, approximately 18 degrees apart as shown by the spacing angle **A2**. In one aspect, the pipe connection ring **300** thus comprises twenty different relief cuts **1535**.

(81) In one aspect, as shown in FIGS. **17** and **18**, an exemplary pipe system **90** comprises the pipe coupling **100**, a first pipe element **110**, and a second pipe element **120**. The pipe coupling **100** comprises an upper housing represented by the coupling segment **200a**, a lower housing represented by the coupling segment **200b**, and the gasket **150**. The coupling segment **200a** and the coupling segment **200b** are held together by nuts and bolts or other fasteners (not shown). In one aspect, the pipe coupling **100** can comprise pipe connection rings **300a,b** sized to fit within respective connection grooves **112,122** defined in the first pipe element **110** and the second pipe element **120** proximate to ends of the first pipe element **110** and the second pipe element **120**. The engagement of the pipe connection rings **300a,b** in the connection grooves **112,122** prevents separation of the first pipe element **110** from the second pipe element **120**, and the seal created by the gasket **150** against outer surfaces **111,121** of the first pipe element **110** and the second pipe element **120** when the pipe coupling **100** is tightened prevents leakage from the joint between the pipe element **110** and the second pipe element **120**.

(82) It may be advantageous for the entire pipe coupling **100** to be able to slide past an end **115** of the pipe element **110** during installation of the pipe coupling **100** as shown in FIGS. **19A-19E**. As shown in FIG. **19C**, for example, being able to slide the entire pipe coupling **100** past the end **115** of the pipe element **110** during installation may facilitate the accurate alignment of the first pipe element **110** with the second pipe element because the pipe coupling **100** need not obscure the joint while still being installed on the first pipe element **110**. Being able to slide the entire pipe coupling **100** past the end **115** of the pipe element **110** during installation also permits one to assemble the pipe coupling **100** before installing it on the pipe elements **110,120**. Moreover, the pipe elements **110,120** and the pipe coupling **100** can be heavy or difficult to position or align, and it is often easier to assemble the pipe coupling **100** as a separate step before installing the pipe coupling **100**

on the pipe elements **110,120**. To facilitate sliding of the entire pipe coupling **100** over the end **115** of the pipe element **110**, the pipe connection rings **300a,b** and the inner surface **202a,b** of the coupling segments **200a,b** may have an inner diameter that is greater than an outer diameter of the pipe elements **110,120** when the pipe coupling is in an assembled but untightened state. The gasket **150** may also have an inner diameter that is greater than an outer diameter of the pipe elements **110,120** when the pipe coupling is in an assembled but untightened state, or the gasket **150** may have one or more portions that extend radially inward past an outer surface of the pipe elements **110,120**.

(83) FIGS. **19A-19E** illustrate the installation of the pipe coupling **100** on the pair of pipe elements **110,120**. The pipe coupling **100** is introduced to a pair of pipe elements **110,120** in the preassembled but untightened position of FIG. **2**. Each pipe element **110,120** may define a connection groove **112,122** proximate to an end of each pipe element **110,120** for alignment with pipe connection rings **300a,b** and to prevent separation of the first pipe element **110** from the second pipe element **120**. In the currently described method, the pipe coupling **100** is axially aligned with the end **115** of the first pipe element **110** as shown in FIG. **19A**. The pipe coupling **100** is installed by placing the end **115** of the first pipe element **110** in a coupling void **105** defined by the pipe coupling **100** and sliding the pipe coupling **100** over the end **115** of the first pipe element **110** as shown in FIG. **19B**. As shown in FIG. **19B**, an inner diameter of the gasket is greater than an outer diameter of the first pipe element **110** and the second pipe element **120**, resulting in the sealing surfaces **526a,b** and the sealing surfaces **532a,b** of the gasket **150** being a small distance away from an outer surface **111** of the first pipe element **110**. The sealing surfaces **526a,b** and **532a,b** may contact the outer surface **111** in other aspects. In one aspect, the entire pipe coupling **100** is slid beyond the connection groove **112** of the first pipe element **110** so that the pipe connection rings **300a,b** and each of the axial ends **205a,b** and **206a,b** have passed the connection groove **112** or at least the end **115** of the first pipe element **110**. In another aspect, it may not be necessary to slide the entire pipe coupling **100** beyond the connection groove **112** or the end **115**. As shown in FIG. **19C**, the second pipe element **120** having an end **125** is introduced in end-facing relationship to the end **115** of the first pipe element **110**.

(84) As shown in FIG. **19D**, when the second pipe element **120** is aligned with the first pipe element **110**, the pipe coupling **100** is moved such that the gasket **150** is positioned around the ends **115,125** of the pipe elements **110,120** and the pipe connection rings **300a,b** are aligned with the connection grooves **112,122**.

(85) As shown in FIG. **19E**, when the coupling segments **200a,b** are clamped down or tightened around the pipe elements **110,120** by tightening the nuts on the bolts or by any other tightening method, the gasket **150** deforms into sealing engagement against the outer surfaces **111,121** of the pipe elements **110,120** and the pipe connection rings **300a,b** sit respectively within the connection grooves **112,122**. A radially innermost portion of the respective inner surface **303a,b** of each of the pipe connection rings **300a,b** can contact a radially innermost portion of a connection groove surface **116,126** that defines a groove bottom of the connection grooves **112,122**.

(86) When the pipe coupling **100** is a flexible-type pipe coupling, a space or gap may remain between the pipe connection ring **300** and the radially innermost portion of the connection groove surface **116,126**. This gap may exist only at specific points or around the full circumference of the pipe connection ring **300**. The pipe connection ring **300** can be configured to compress only to a certain point at which an inner diameter of the pipe connection ring **300** is greater than the outer diameter of the radially innermost portion of the connection groove surface **116,126**. The gap allows or the gaps allow a limited amount of freedom for the pipe elements **110,120** to move within the pipe coupling **100**, which may prove advantageous for some installations including those in earthquake-prone geographic areas. If the pipe connection ring **300** is allowed to compress further, then even complete tightening of the pipe coupling **100** will not cause the pipe connection ring to fully compress. In that event, the pipe coupling segments **200a,b** will deform uniformly and leave

no gap between the pipe connection ring and the radially innermost portion of the connection groove surface **116,126**. This results in a rigid-type pipe coupling, in which the pipe elements **110,120** have no clearance with which to move within the pipe coupling **100**, which may prove advantageous for some installations including those in which the fluid inside the pipe elements **110,120** is at high pressure. An example of a rigid-type configuration is illustrated in FIGS. **19A-19E**.

(87) An axial end surface such as the first axial end surface **301a,b** or the second axial end surface **302a,b** of the pipe connection rings **300a,b** may contact an axially outermost portion of the connection groove surface **116,126** (i.e., axially outermost meaning closest to the ends **115,125**) that defines a sidewall of the connection grooves **112,122**. The connection rings **300a,b** may not contact the sidewalls of the connection grooves **112,122** in other aspects until an internal or external force, such as fluid pressure with the pipe system **90**, acts to separate the pipe elements **110,120**, and the connection rings **300a,b** thereby engage the sidewalls and groove bottoms to prevent this separation. In some aspects, a gap can remain between the transition portions **272,274,276,278** of the inner surface **202a,b** of each coupling segment **200a,b** and a radially outermost portion of the outer surfaces **111,121** of the pipe elements **110,120** upon clamping down the coupling segments **200a,b**, or the transition portions **272,274,276,278** of the inner surface **202a,b** of each coupling segment **200a,b** can come into partial or complete contact with the outer surfaces **111,121** of the pipe elements **110,120** in other aspects.

(88) When the pipe coupling **100** is aligned on the joint of the pipe elements **110,120** as shown in FIG. **19E**, the fastening devices **290a,b** are used to draw the coupling segments **200a,b** together. In one aspect, this takes the form of fastening devices **290a,b** compressing the fastener pads **221,222** toward each other. In one aspect, the pipe coupling **100** is rigid and includes no deflection of the coupling segments **200a,b**. In another aspect, the coupling segment is deformable between a non-assembled state and an assembled and tightened state, each coupling segment **200a,b** of the pipe coupling **100** deflecting at least slightly when the pipe coupling **100** is tightened. In various aspects, when the coupling segments **200a,b** travel toward each other and deform under the tightening of the fastening devices **290**, the gasket **150** is deformed in accord therewith. In some embodiments, a rigid or semi-rigid gasket **150** may be included. The process for accommodating such a material may be altered from that described herein; for example and without limitation, however, the gasket **150** can comprise the deformation groove **517** to allow a place for material to flow upon deformation of the gasket **150**.

(89) Several features of the gasket **150** ease installation as described. Friction can cause installation of rubber gaskets to bind against outer surfaces **111,121** of pipe elements **110,120**. With reference to the method shown FIGS. **19A-19E**, axially outer drafted edges and axially inner drafted edges of the gasket **150** are both drafted to ease the pipe elements **110,120** into the coupling void **105**. Axially outer drafted edges and axially inner drafted edges also help to prevent rollover of the sealing ridges **525a,b** of the gasket **150** while sliding the pipe coupling **100** on and off, as the drafted profiles are less likely to bind against the outer surfaces **111,121** of the pipe elements **110,120**. Additionally, sealing surfaces **526a,b** are substantially parallel to the outer surfaces **111,121** of the pipe elements **110,120** even when the gasket **150** is not tightened against the pipe elements **110,120**. Additionally, the axially outward position of the central rib **530** with respect to the sealing ridges **525a,b**, prevents the central rib **530** from obstructing the installation of the pipe coupling **100** before deformation of the gasket **150**. When the coupling segments **200a,b** are clamped down, the gasket **150** deforms, and the central rib **530** contacts the outer surfaces **111,121** of the pipe elements **110,120**. This configuration allows the gasket **150** to slide onto the pipe elements **110,120** without biasing the sliding in one direction and prevents binding of the sealing ridges **525a,b** during installation. These features prevent the gasket **150** from rolling over when the pipe coupling **100** is installed on the pipe elements **110,120** and allows the gasket **150** to be properly placed over the joint between the pipe elements **110,120** for proper sealing, among other

advantages. When the gasket **150** is properly aligned over the joint and engaged against the outer surfaces **111,121** of the pipe elements **110,120**, each of the sealing members **557a,b** may contact the outer surfaces **111,121**.

(90) FIG. **20** again shows the pipe coupling **100** assembled and tightened around and connecting the pipe elements **110,120**. The pipe elements **110,120** respectively comprise shoulders **118,128** defining sidewalls **119,129** instead of having connection grooves **112,122**. One of the axial end surfaces **301a,b** or the axial end surfaces **302a,b** of each of the pipe connection rings **300a,b** may contact one of the sidewalls **119,129** of the shoulders **118,128** to engage the pipe elements **110,120** when an internal or external force, such as fluid pressure with the pipe system **90**, acts to separate the pipe elements **110,120**. This contact prevents separation of either pipe element **110,120** from the pipe coupling **100** or from each other once the pipe coupling **100** is tightened. The pipe coupling **100** disclosed herein could be used with plain-end pipe elements as well that do not incorporate a connection groove **112,122** or a shoulder **118,128**.

(91) In one aspect, the pipe elements **110,120** are approximately the same diameter, although non-uniform diameter pipe elements may be joined in various embodiments. In one aspect, each of the pipe elements **110,120** has ends **115,125** that define a diameter equal to a diameter of the respective pipe elements **110,120**. In another aspect, the ends **115,125** can be flared slightly. Connection grooves **112,122** can be formed using one or more of at least two methods: rolling and machining. If the connection grooves **112,122** are machined, the pipe elements **110,120** are less likely to have flares on the ends **115,125** and are more likely to resemble the geometry shown. If the connection grooves **112,122** are rolled, however, the pipe elements **110,120** are more likely to have ends **115,125** which are flared, though rolled pipe elements **110,120** may not have flares in other aspects. As such, the pipe coupling **100** can be made to accommodate the potential flaring of ends **115,125**. When the ends **115,125** are flared, the pipe coupling **100** can be sized to fit over the largest possible flare of the ends **115,125** based on standard tolerancing for creating the connection grooves **112,122**.

(92) In one aspect, the gasket **150** comprises the geometry shown in FIGS. **21** through **23** including a pair of sealing ribs **520a,b** and a plurality of tabs **550** distributed circumferentially about and protruding from the central rib **530**. In cross-section, the gasket **150** defines the overall height **570** measured from the radially outer surface **515** to the sealing surfaces **526a,b** of the sealing ridges **525a,b**. The overall height **570** can be reduced as shown—in comparison to the overall height **570** of the gasket **150** shown in FIG. **2**. In one aspect, reducing the overall height **570** causes the gasket **150** to protrude radially outward less with respect to a given inner diameter of the gasket **150**, giving the gasket **150** a lower profile to allow the coupling segments **200a,b** and the pipe connection rings **300** to have smaller diameters to closer match the diameter of the pipes and thereby lower the amount of tightening necessary to engage the pipe elements **110,120**.

(93) In another aspect, reducing the overall height **570** causes the gasket **150** to protrude radially inward less than the distance shown in FIG. **2** relative to the respective inner surface **202a,b** of the coupling segments **200a,b** or the respective inner surfaces of the pipe connection rings **300a,b**. By reducing the overall height **570** of the gasket **150** as shown, with or without the inclusion of the tabs **550**, the pipe connection rings **300a,b** of the pipe coupling **100** can be made to align with the connection grooves **112,122** or on the outside of the shoulders **118,128**, at a point in the installation process before the gasket **150** contacts the pipe elements **110,120** or before the gasket **150** compresses significantly. In one aspect, a user may choose to rely on tactile sensation (i.e., may rely on “feel”) in addition to or instead of visual clues to confirm proper alignment of the pipe coupling **100** with the pipe elements **110,120**. In one aspect, tactile confirmation that both of the pipe connection rings **300a,b** are simultaneously engaged in the connection grooves **112,122** or on the outside of the shoulders **118,128** will be more difficult as the gasket **150** compresses. Delaying compression of the gasket **150** can be achieved, for example and without limitation, by reducing the overall height **570** of the gasket **150**.

(94) In one aspect, as shown, each of the tabs **550** when bent individually and the plurality of tabs **550** when bent simultaneously are easily bendable in an axial direction so as not to interfere with installation of the gasket completely over the ends **115,125** of either of the pipe elements **110,120**. In one aspect, an innermost diameter of the gasket **150**, as measured to a radially inward facing portion of a tab surface **551**, is less than the outer diameter of the pipe elements **110,120**. When sliding a gasket such as the gasket **150** shown in FIG. **23** completely over the end **115,125** of either of the pipe elements **110,120**, one of the respective ends **115,125** of the pipe elements **110,120** contacts an axially outward facing portion of the tab surface **551** of each of the tabs **550**, causing each of the tabs **550** to bend. When the gasket **150** is slid back in the other direction, including when the pipe elements **110,120** are positioned end-to-end but with a slight gap, the tabs **550** will unbend and extend into the gap between the pipe elements **110,120**. In such aspect, axially outward facing portions of the tabs **550** face axially outward facing portions of the respective ends **115,125** of the pipe elements **110,120**. The tabs **550** can thus engage the ends **115,125** to separate the end **115** from the end **125** and to provide proper spacing for the pipe connection rings **300a,b** to align with the connection grooves **112,122**. The gap created between the end **115** and the end **125** by the tabs **550** can be at least equal to an axial thickness of the tab **550** as viewed in cross-section such as in FIG. **23**. The geometry of the pipe elements **110,120** and or features of the pipe coupling **100** can be adjusted as necessary to accommodate the axial thickness of the tab **550** including additional space for any manufacturing tolerances.

(95) In one aspect, each of the tabs **550** bends easily when the gasket **150** is slid completely over the end **115,125** of one of the pipe elements **110,120** because each of the tabs **550** is rounded on each axially outward edge. In another aspect, each of the tabs **550** bends easily in the described condition because each tab **550** is separate and distinct from every other tab **550** and therefore bending of each of the tabs **550** in an axial direction does not require the simultaneously stretching or deformation of any material adjacent to the tab **550**—such as would be the case were the geometry of the tab **550** such as that shown in FIG. **23** to continue circumferentially around the gasket **150**. In one aspect, when the gasket **150** is viewed along its axis, each of the tabs **550** also tapers from the central rib **530**, defining taper tangent lines **555a,b** extending toward a radially inward facing portion of the tab surface **551** and angled with respect to a line **557** extending through a center of the gasket. With this taper, the cross-sectional area of the tab **550** effectively decreases towards the radially inward facing portion of the tab surface **551** and therefore less force is required to cause the tab **550** to bend at the radially inward facing portion of the tab surface than is required to cause the tab **550** to bend proximate to its connection to the central rib **530**. In addition, because the taper on each tab **550** causes less surface area of each of the tabs **550** to contact the pipe elements **110,120**, the frictional forces created when sliding the gasket **150** completely over either of the respective ends **115,125** of the pipe elements **110,120** are reduced. In one aspect, for example and without limitation, the circumferential length of the gasket including tabs **550** can be less than the circumferential length of the gasket not including tabs. In one aspect, a total of six tabs **550** are evenly spaced circumferentially around the gasket **150**, where each tab **550** is spaced apart from each adjacent tab by an angle of 60° . In another aspect, a smaller quantity or a greater quantity of tabs **550** are spaced circumferentially around the gasket **150**, and the tabs **550** need not be spaced evenly. In one aspect, an intersection between the tab surface **551** and a radially inward facing surface of the gasket **150** includes a radius or other fillet, reducing or eliminating any stress concentration at the intersection.

(96) In one aspect, each of the sealing ribs **520a,b** extends substantially radially inwardly and increases in thickness from a radially outside position to a radially inside position. In one aspect, each of the sealing ribs **520a,b** defines an axially outer surface **521a,b** extending from the radially outer surface **515** to an end of an axially outer drafted surface **522a,b**, respectively. Each of the axially outer surfaces **521a,b** is angled or slanted with respect to a radial direction of the gasket **150**. The angle of each of the axially outer surfaces **521a,b** is consistent around the circumference

of the gasket **150**, so that each of the axially outer surfaces **521a,b** is shaped as a truncated cone. In one aspect as shown, each of the axially outer surfaces **521a,b** is angled, for example and without limitation, between about 19° and about 22° degrees with respect to a radial direction.

(97) Each of the axially outer drafted surfaces **522a,b** extends from an axially outermost edge of the gasket **150** to the corresponding sealing surface **526a,b**. In one aspect, each of the axially outer drafted surfaces **522a,b** may be rounded, slanted, or have any one of various other shapes in cross-section. In one aspect, such cross-sectional shapes translate to the three-dimensional shape of a cone or a paraboloid. Such shapes are truncated, as a full cone or paraboloid would not typically allow insertion of pipe elements in the gasket **150**. In one aspect as shown, each of the axially outer drafted surfaces **522a,b** is angled, for example and without limitation, between about 27° and about 28° with respect to a radial direction. Extending from each of the sealing surfaces **526a,b** is an axially inner drafted surface **527a,b**. In one aspect, each of the axially inner drafted surfaces **527a,b** may be rounded, slanted, or have any one of various other shapes in cross-section. In one aspect, such cross-sectional shapes translate to the three-dimensional shape of a cone or a paraboloid. Such shapes are truncated, as a full cone or paraboloid would not typically allow insertion of pipe elements in the gasket **150**. Each of the axially inner drafted surfaces **527a,b** defines the termination of the sealing ridge **525a,b** along an axially inward direction.

(98) In another aspect, the gasket **150** comprises the geometry shown in FIGS. **24** through **26** including the pair of sealing ribs **520a,b** and the plurality of tabs **550** distributed circumferentially about and protruding from the central rib **530**. In one aspect, each of the tabs **550** is positioned axially between the sealing member **557a** and the sealing member **557b**. In one aspect, the overall height **570** can be further reduced as shown—in comparison to the overall height **570** of the gasket **150** shown in FIG. **2** and in comparison to the overall height **570** of the gasket **150** shown in FIG. **23**. In one aspect, reducing the overall height **570** causes the gasket **150** to protrude radially outward less with respect to a given inner diameter of the gasket **150**, giving the gasket **150** a lower profile to allow the coupling segments **200a,b** and the pipe connection rings **300** to have smaller diameters to closer match the diameter of the pipes and thereby lower the amount of tightening necessary to engage the pipe elements **110,120**.

(99) In another aspect, reducing the overall height **570** causes the gasket **150** to protrude radially inward by less than the distances shown in FIGS. **2** and **23** relative to the respective inner surface **202a,b** of the coupling segments **200a,b** or the respective inner surfaces of the pipe connection rings **300a,b**.

(100) In one aspect, a method of installing the pipe coupling **100** in the pipe system **90** comprises connecting the first pipe element **110** to the second pipe element **120**. In one aspect, this method comprises sliding the assembled pipe coupling **100** over the end **115** of the first pipe element **110**, the pipe coupling **100** comprising a coupling segment **200** and a pipe connection ring **300**, the coupling segment **200** having an inner surface **202** defining a ring groove **250**, the pipe connection ring **300** disposed within the ring groove **250**, the pipe connection ring **300** being a closed ring. In one aspect, the method further comprises compressing the pipe connection ring **300** in a radial direction to engage the first pipe element **110**.

(101) In one aspect, compressing the pipe connection ring **300** in a radial direction comprises engaging a connection groove **112,122** defined in the first pipe element **110** proximate to an end **115** of the first pipe element **110** by compressing the pipe connection ring **300** such that an overall inner diameter equaling twice the overall inner radius **1091** of the pipe connection ring **300** becomes smaller than an outer diameter of the first pipe element **110**. In one aspect, compressing the pipe connection ring **300** further comprises tightening a fastening device **290** of the pipe coupling **100** to bring the coupling segment **200** closer to the first pipe element **110**.

(102) One should note that conditional language, such as, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not

include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more particular embodiments or that one or more particular embodiments necessarily include logic for deciding, with or without user input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular embodiment.

(103) It should be emphasized that the above-described embodiments are merely possible examples of implementations, merely set forth for a clear understanding of the principles of the present disclosure. Any process descriptions or blocks in flow diagrams should be understood as representing modules, segments, or portions of code which include one or more executable instructions for implementing specific logical functions or steps in the process, and alternate implementations are included in which functions may not be included or executed at all, may be executed out of order from that shown or discussed, including substantially concurrently or in reverse order, depending on the functionality involved, as would be understood by those reasonably skilled in the art of the present disclosure. Many variations and modifications may be made to the above-described embodiment(s) without departing substantially from the spirit and principles of the present disclosure. Further, the scope of the present disclosure is intended to cover any and all combinations and sub-combinations of all elements, features, and aspects discussed above. All such modifications and variations are intended to be included herein within the scope of the present disclosure, and all possible claims to individual aspects or combinations of elements or steps are intended to be supported by the present disclosure.

Claims

1. A pipe connection ring comprising: a plurality of segments; and a plurality of deformable links, each of the plurality of links connecting adjacent segments of the plurality of segments, the segments and links alternating about a circumference of the pipe connection ring, each pair of the adjacent segments and a corresponding link of the plurality of links connecting the adjacent segments and together with the adjacent segments defining a relief between the adjacent segments, each of the plurality of links configured to allow movement of the adjacent segments upon deformation of the link connecting the adjacent segments; wherein the pipe connection ring defines a first axial end surface, a second axial end surface distal from the first axial end surface, an inner surface, and an outer surface distal from the inner surface; wherein an axial ring thickness of each of the plurality of segments of the pipe connection ring remains constant before and after compression of the pipe connection ring, wherein the plurality of segments have a constant radial ring thickness, and wherein at least one of a first end and a second end of one of the plurality of segments is angled by a slant angle with respect to a tangent line defined by a circumferential axis of the pipe connection ring.
2. The pipe connection ring of claim 1, wherein the pipe connection ring is a closed ring.
3. The pipe connection ring of claim 1, wherein each of the inner surface and the outer surface define a radius, the radius of the outer surface being greater than the radius of the inner surface.
4. The pipe connection ring of claim 1, wherein each of the inner surface, the outer surface, and surfaces of the relief are orthogonal to each of the first axial end surface and the second axial end surface.
5. The pipe connection ring of claim 1, wherein the relief defines a relief angle between a pair of surfaces defining the relief, the relief angle measuring greater than zero degrees.
6. The pipe connection ring of claim 1, wherein each link of the plurality of links comprises a pair of legs joined by a hairpin bend, a first leg of the pair of legs joined to a first segment of a corresponding pair of adjacent segments and a second leg of the pair of legs joined to a second segment of the pair of adjacent segments.
7. A pipe connection ring comprising: a plurality of segments; and a plurality of deformable links,

each of the plurality of links connecting adjacent segments of the plurality of segments, the segments and links alternating about a circumference of the pipe connection ring, each pair of the adjacent segments and a corresponding link of the plurality of links connecting the adjacent segments and together with the adjacent segments defining a relief between the adjacent segments, each of the plurality of links configured to allow movement of the adjacent segments upon deformation of the link connecting the adjacent segments; wherein the pipe connection ring defines a first axial end surface, a second axial end surface distal from the first axial end surface, an inner surface, and an outer surface distal from the inner surface; wherein an axial ring thickness of each of the plurality of segments of the pipe connection ring remains constant before and after compression of the pipe connection ring, wherein the plurality of segments have a constant radial ring thickness, and wherein a first segment of a first pair of adjacent segments is of a different size than a second segment of the first pair of adjacent segments.

8. The pipe connection ring of claim 7, wherein a spacing angle defined by the first pair of adjacent segments measures less than or equal to approximately 22.5 degrees.

9. A pipe connection ring, comprising: a plurality of segments; and a plurality of deformable links, each of the plurality of links connecting adjacent segments of the plurality of segments, the segments and links alternating about a circumference of the pipe connection ring, each pair of the adjacent segments and a corresponding link of the plurality of links connecting the adjacent segments and together with the adjacent segments defining a relief between the adjacent segments, each of the plurality of links configured to allow movement of the adjacent segments upon deformation of the link connecting the adjacent segments; wherein the pipe connection ring defines a first axial end surface, a second axial end surface distal from the first axial end surface, an inner surface, and an outer surface distal from the inner surface; and wherein an axial ring thickness of each of the plurality of segments of the pipe connection ring remains constant before and after compression of the pipe connection ring, wherein during compression of the pipe connection ring a first segment of a first pair of adjacent segments rotates with respect to a second segment of the first pair of adjacent segments.

10. A pipe coupling comprising: a coupling segment defining an inner surface defining a first ring groove proximate to a first axial end of the coupling segment, the inner surface of the coupling segment comprising a transition portion on each side of the first ring groove, the first ring groove defining a groove bottom surface positioned radially outward from each of the transition portions; and the pipe connection ring of claim 1 disposed within the first ring groove.

11. The pipe coupling of claim 10, wherein the pipe connection ring is a closed ring.

12. The pipe coupling of claim 10, wherein the inner surface of the coupling segment further defines a second ring groove proximate to a second axial end of each of the coupling segment, the inner surface of the coupling segment comprising a transition portion on each side of the second ring groove, the second ring groove defining a groove bottom surface positioned radially outward from each of the transition portions proximate to the second ring groove, the pipe coupling further comprising a second pipe connection ring disposed within the second ring groove, the second pipe connection ring comprising: a plurality of segments; and a plurality of deformable links, each of the plurality of links of the second pipe connection ring connecting adjacent segments of the plurality of segments of the second pipe connection ring, the segments and links of the second pipe connection ring alternating about a circumference of the second pipe connection ring, each pair of the adjacent segments and a link of the plurality of links of the second pipe connection ring connecting the adjacent segments and together with the adjacent segments defining a relief between the adjacent segments of the second pipe connection ring, each of the plurality of links of the second pipe connection ring configured to allow movement of the adjacent segments of the second pipe connection ring upon deformation of the link connecting the adjacent segments of the second pipe connection ring; wherein the second pipe connection ring defines a first axial end surface, a second axial end surface distal from the first axial end surface of the second pipe connection ring,

an inner surface, and an outer surface distal from the inner surface of the second pipe connection ring.

13. A pipe system comprising: a first pipe element; a second pipe element; and the pipe coupling of claim 10, the pipe coupling connecting an end of the second pipe element to an end of the first pipe element.

14. A pipe connection ring comprising: a plurality of segments; and a plurality of deformable links, each of the plurality of links connecting adjacent segments of the plurality of segments, the segments and links alternating about a circumference of the pipe connection ring, each pair of the adjacent segments and a corresponding link of the plurality of links connecting the adjacent segments and together with the adjacent segments defining a relief between the adjacent segments, each of the plurality of links configured to allow movement of the adjacent segments upon deformation of the link connecting the adjacent segments; wherein the pipe connection ring defines a first axial end surface, a second axial end surface distal from the first axial end surface, an inner surface, an outer surface distal from the inner surface, and a central axis, each of the first axial end surface and the second axial end surface being angled 90 degrees with respect to the central axis, wherein the plurality of segments have a constant radial ring thickness, and wherein during compression of the pipe connection ring a first segment of a first pair of adjacent segments rotates with respect to a second segment of the first pair of adjacent segments.

15. The pipe connection ring of claim 14, wherein the pipe connection ring is a closed ring.

16. The pipe connection ring of claim 14, wherein each of the inner surface, the outer surface, and surfaces of the relief are orthogonal to each of the first axial end surface and the second axial end surface.

17. The pipe connection ring of claim 14, wherein the relief defines a relief angle between a pair of surfaces defining the relief, the relief angle measuring greater than zero degrees.

18. The pipe connection ring of claim 14, wherein a first segment of a first pair of adjacent segments is of a different size than a second segment of the first pair of adjacent segments.

19. The pipe connection ring of claim 14, wherein each link of the plurality of links comprises a pair of legs joined by a hairpin bend, a first leg of the pair of legs joined to a first segment of a corresponding pair of adjacent segments and a second leg of the pair of legs joined to a second segment of the pair of adjacent segments.

20. A pipe coupling comprising: a coupling segment defining an inner surface defining a first ring groove proximate to a first axial end of the coupling segment, the inner surface of the coupling segment comprising a transition portion on each side of the first ring groove, the first ring groove defining a groove bottom surface positioned radially outward from each of the transition portions; and the pipe connection ring of claim 14 disposed within the first ring groove.

21. The pipe coupling of claim 20, wherein the pipe connection ring is a closed ring.

22. The pipe coupling of claim 20, wherein the inner surface of the coupling segment further defines a second ring groove proximate to a second axial end of each of the coupling segment, the inner surface of the coupling segment comprising a transition portion on each side of the second ring groove, the second ring groove defining a groove bottom surface positioned radially outward from each of the transition portions proximate to the second ring groove, the pipe coupling further comprising a second pipe connection ring disposed within the second ring groove, the second pipe connection ring comprising: a plurality of segments; and a plurality of deformable links, each of the plurality of links of the second pipe connection ring connecting adjacent segments of the plurality of segments of the second pipe connection ring, the segments and links of the second pipe connection ring alternating about a circumference of the second pipe connection ring, each pair of the adjacent segments and a link of the plurality of links of the second pipe connection ring connecting the adjacent segments and together with the adjacent segments defining a relief between the adjacent segments of the second pipe connection ring, each of the plurality of links of the second pipe connection ring configured to allow movement of the adjacent segments of the second

pipe connection ring upon deformation of the link connecting the adjacent segments of the second pipe connection ring; wherein the second pipe connection ring defines a first axial end surface, a second axial end surface distal from the first axial end surface of the second pipe connection ring, an inner surface, and an outer surface distal from the inner surface of the second pipe connection ring.

23. A pipe system comprising: a first pipe element; a second pipe element; and the pipe coupling of claim 20, the pipe coupling connecting an end of the second pipe element to an end of the first pipe element.
